

Module - 1: Electrochemistry

Electrode Potential (E)

When metal rod dipped in their salt solution potential is developed are called electrode potential.

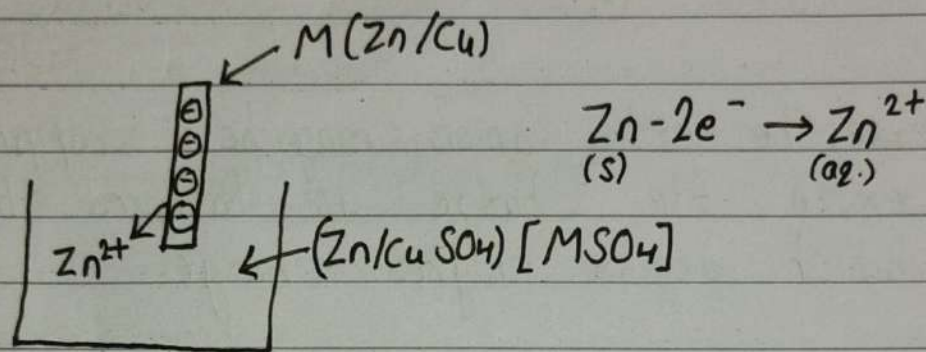
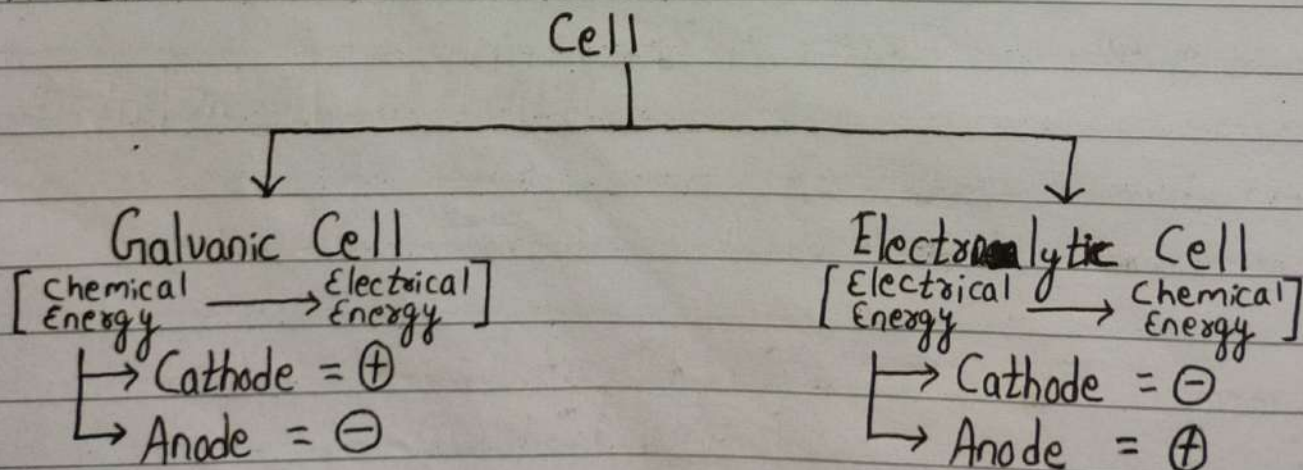


Fig: Electrode

Standard Electrode Potential (E°)

At 298 K and 1 Atm. pressure, metal rod dipped in their salt solution one molar conc., then potential is developed are called standard electrode potential.

Cell



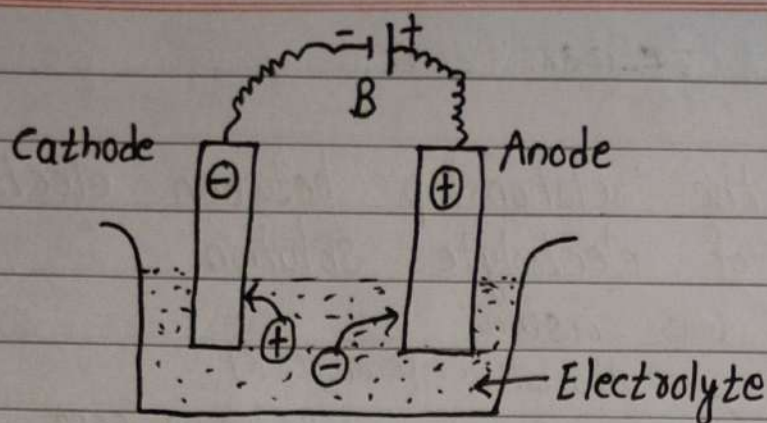


Fig: Electrolytic Cell

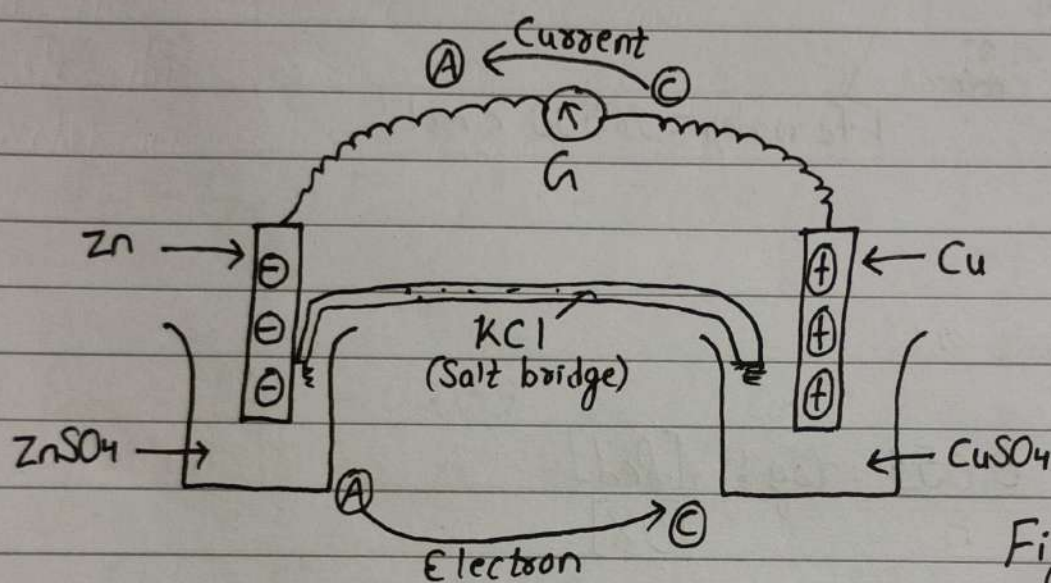
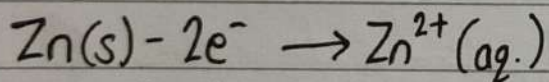
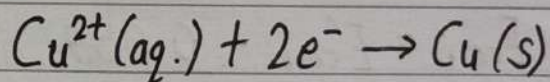


Fig: Galvanic Cell



Oxidation

Anode [OA]



Reduction

Cathode [RC]

Half-cell

Cell Representation: $\text{Zn(s)} | \text{Zn}^{2+}(\text{aq.}) || \text{Cu}^{2+}(\text{aq.}) | \text{Cu(s)}$

$$E_{\text{cell}} = E_{\text{cathode}} - E_{\text{anode}}$$

$$E_{\text{cell}}^{\circ} = E_{\text{cathode}}^{\circ} - E_{\text{anode}}^{\circ}$$

Electro-chemical Series: $\text{Cu}^{2+} = +0.34\text{V}$
 $\text{Zn}^{2+} = -0.76\text{V}$

Nernst Equation

This equation explains the relationship between electrode potential and conc. of electrolyte solution.

$$E = E^{\circ} - \frac{2.303 RT}{nF} \log_{10} \frac{[\text{Red.}]}{[\text{OX}]}$$

Annotations:

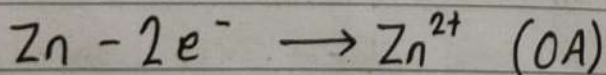
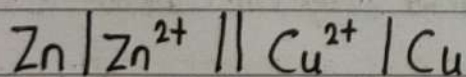
- E → Electrode Potential
- E° → Standard E.P.
- R → Gas constant
- T → Absolute Temperature (K)
- n → No. of e^- on electrode
- F → 1 Faraday = 96500 C mol^{-1}
- $[\text{Red.}]$ → Reducing conc.
- $[\text{OX}]$ → Oxidising conc.

At 298 K,

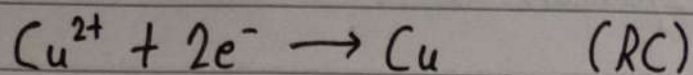
$$E_{\text{cell}} = E^{\circ}_{\text{cell}} - \frac{0.059}{n} \log_{10} \frac{[\text{Red.}]}{[\text{OX}]}$$

$$E_{\text{cell}} = E^{\circ}_{\text{cell}} - \frac{0.059}{n} \log_{10} \frac{1}{[M^{n+}]}$$

EMF of Galvanic Cell



$$E_{\text{Zn}} = E^{\circ}_{\text{Zn}} - \frac{0.059}{n} \log \frac{1}{[\text{Zn}^{2+}]} \quad \text{--- (1)}$$



$$E_{\text{Cu}} = E_{\text{Cu}}^{\circ} - \frac{0.059}{n} \log \frac{1}{[\text{Cu}^{2+}]} \quad \text{--- (2)}$$

$$\text{EMF } (E_{\text{cell}} / E_{\text{cell}}^{\circ}) = E_{\text{C}} - E_{\text{A}}$$

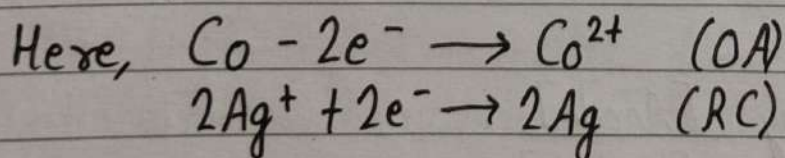
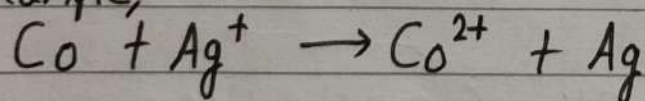
$$\text{Eq. (2) - (1),}$$

$$E_{\text{Cu}} - E_{\text{Zn}} = E_{\text{Cu}}^{\circ} - E_{\text{Zn}}^{\circ} - \frac{0.059}{n} \left(\log \frac{1}{[\text{Cu}^{2+}]} - \log \frac{1}{[\text{Zn}^{2+}]} \right)$$

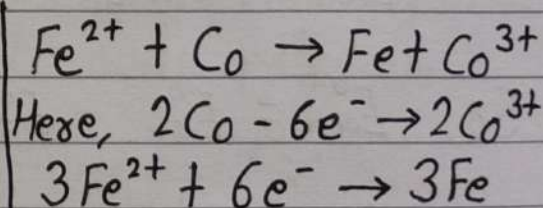
$$E_{\text{cell}} = E_{\text{cell}}^{\circ} - \frac{0.059}{n} \log \frac{1/[\text{Cu}^{2+}]}{1/[\text{Zn}^{2+}]}$$

$$E_{\text{cell}} = E_{\text{cell}}^{\circ} - \frac{0.059}{n} \log \left(\frac{[\text{Zn}^{2+}]}{[\text{Cu}^{2+}]} \right)$$

For example,



$$E_{\text{cell}} = E_{\text{cell}}^{\circ} - \frac{0.059}{2} \log \frac{[\text{Co}^{2+}]}{[\text{Ag}^+]^2}$$

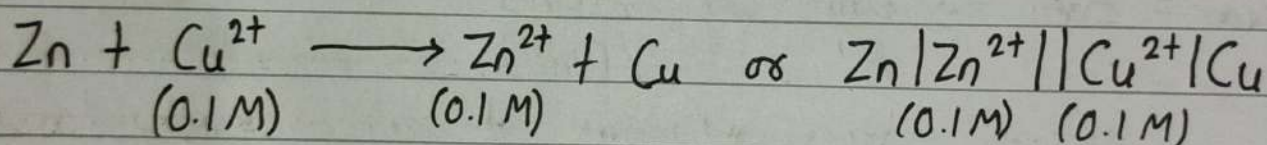


$$E_{\text{cell}} = E_{\text{cell}}^{\circ} - \frac{0.059}{6} \log \frac{[\text{Co}^{3+}]^2}{[\text{Fe}^{2+}]^3}$$

Applications of Nernst Equation

1. It is used to calculate ion concentration.
2. It is used in pH measurement.
3. It can be used in potentiometric titration and solubility product.
4. It can be used ^{to determine} ~~in~~ ionization constant and degree of dissociation of weak electrolyte.

Q) Calculate EMF of cell



Solⁿ:

$$\text{Given, } E_{\text{Zn}^{2+}/\text{Zn}}^{\circ} = -0.76 \text{ V}$$

$$E_{\text{Cu}^{2+}/\text{Cu}}^{\circ} = +0.34 \text{ V}$$

$$\text{So, } E_{\text{cell}}^{\circ} = E_{\text{Cu}}^{\circ} - E_{\text{Zn}}^{\circ} = 0.34 - (-0.76) = 1.10 \text{ V}$$

$$\begin{aligned} \therefore E_{\text{cell}} &= E_{\text{cell}}^{\circ} - \frac{0.059}{n} \log \frac{[\text{Anode}]}{[\text{Cathode}]} \\ &= 1.10 - \frac{0.059}{2} \log \frac{[\text{Zn}^{2+}]}{[\text{Cu}^{2+}]} \\ &= 1.10 - \frac{0.059}{2} \log \frac{0.1}{0.1} \end{aligned}$$

$$\therefore E_{\text{cell}} = +1.10 \text{ V}$$